CHARLES **DU FAY**

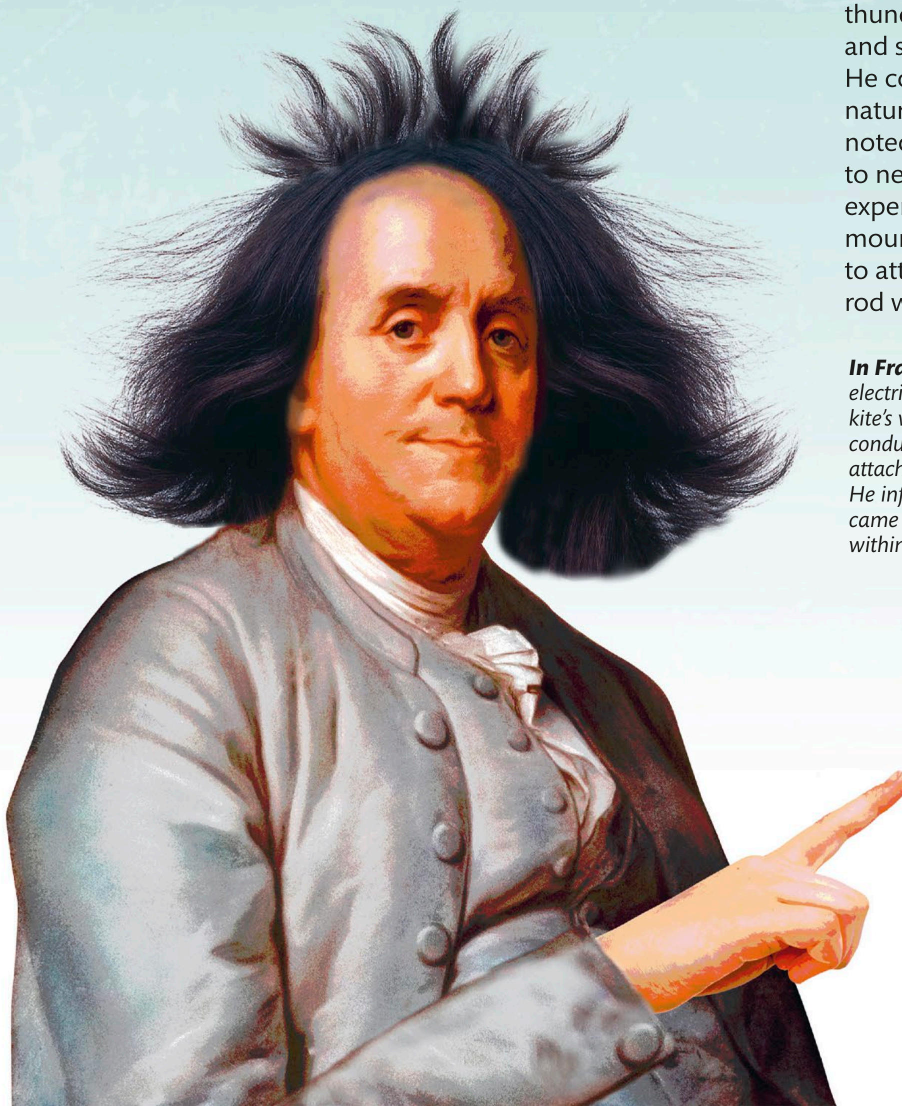
French chemist Charles François de Cisternay du Fay made early contributions to the scientific study of electrically charged objects and paved the way for the work of Benjamin Franklin.

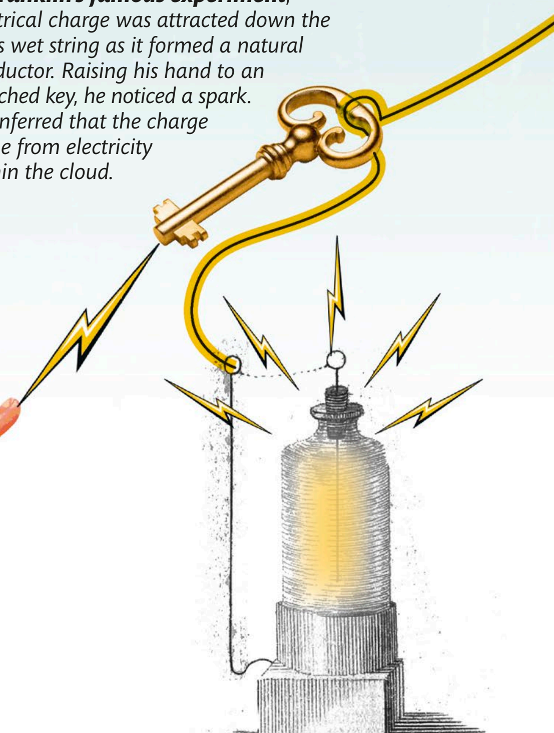
A former soldier, du Fay (1698–1739) gained renown in the 1730s for his theories concerning the properties of electric charge. Having noticed differences in the way charge behaves, he proposed there were two different types of electricity: *vitreous* and *resinous*. These were later named by Franklin as “positive” and “negative.” Du Fay also observed that like charges repel, whereas opposing charges attract, and recorded the difference between conductors and insulators.

a “fluid” that came in two kinds, one that attracted and one that repelled—was incorrect. Instead, he argued that there was only one kind of electricity, which had either a positive or a negative charge. He further proposed that electricity could be neither created nor destroyed, but only transferred from one object to another, while its total amount of charge remained the same. This idea later became known as the law of the conservation of charge.

Natural electricity

Franklin also saw a correlation between thunder and lightning and the bangs and sparks generated by his experiments. He concluded that lightning must be natural electricity. Having previously noted that electricity was attracted to needle points, he proposed an experiment in which an iron rod was mounted on a large insulated box to attract a lightning bolt. When the rod was struck by lightning, it produced





***In Franklin’s famous experiment**, electrical charge was attracted down the kite’s wet string as it formed a natural conductor. Raising his hand to an attached key, he noticed a spark. He inferred that the charge came from electricity within the cloud.*